divide-and-conquer)

- Graph algorithms, tree traversals

6. Theory of computation

- Regular expressions, finite automata, context-free grammars, Pushdown automata.
- Turning machines, undecidability

7. Compiler Design

- Lexical analysis, parsing, syntax-directed translation, runtime environments, intermediate code, local/global optimization.

8. Operating System

- processes, threads, scheduling, synchronization, deadlock, memory management, Virtual memory, file ~~managemen~~ systems.

9. Databases

- ER model, relational model, SQL, normalization, transactions, concurrency control, indexing (B/B+ trees), file Organization

10. Computer Networks

- OSI/TCP-IP stack, LAN Technologies, switching, routing, IP addressing, TCP/UDP, application layer protocols.